

Feature Engineering & Outlier Detection

Practice Problems Code Solutions

Algorithm Engineering Report

Overview

When preprocessing data for algorithmic trading models or structuring scraped web data, outlier detection prevents anomalous spikes from skewing moving averages or predictive signals. This document provides the programmatic solutions to the requested practice problems. The methods implemented in Python include Standard Deviation (Z-Score) analysis, Interquartile Range (IQR) filtering, and Winsorization limits.

1 Problem 1: Z-Score Method (Normal Data)

Dataset (Age): [18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 26, 27, 28, 30, 32, 35, 40, 45, 70]

Python Code

```
1 import numpy as np
2
3 d1 = np.array([18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 26, 27, 28,
4               30, 32, 35, 40, 45, 70])
5 mean1 = np.mean(d1)
6 std1 = np.std(d1, ddof=1) # Sample standard deviation
7
8 # Calculate Z-Scores
9 z_scores1 = (d1 - mean1) / std1
10
11 # Identify and remove outliers (|Z| > 3)
12 outliers1 = d1[np.abs(z_scores1) > 3]
13 d1_clean = d1[np.abs(z_scores1) <= 3]
```

Results

- **Mean (μ):** 28.40
- **Standard Deviation (σ):** 12.13
- **Outliers Identified:** [70]
- **New Dataset Length:** 19 (After removing the outlier)

2 Problem 2: IQR Method (Skewed Data)

Dataset (Monthly salaries): [25, 28, 30, 32, 32, 33, 35, 35, 36, 38, 40, 42, 45, 48, 50, 55, 60, 65, 70, 80, 120, 150, 200, 250, 500]

Python Code

```
1 d2 = np.array([25, 28, 30, 32, 32, 33, 35, 35, 36, 38, 40, 42, 45, 48,
2               50,               55, 60, 65, 70, 80, 120, 150, 200, 250, 500])
3
4 # Calculate Quartiles
5 q1_2 = np.percentile(d2, 25)
6 q3_2 = np.percentile(d2, 75)
7 iqr2 = q3_2 - q1_2
8
9 # Determine Bounds
10 lb2 = q1_2 - 1.5 * iqr2
11 ub2 = q3_2 + 1.5 * iqr2
12
13 # Identify Outliers and Cap
14 outliers2 = d2[(d2 < lb2) | (d2 > ub2)]
15 d2_capped = np.clip(d2, lb2, ub2)
```

Results

- **Q1 (25th Percentile):** 35.00
- **Q3 (75th Percentile):** 70.00
- **IQR:** 35.00
- **Lower Bound:** -17.50 — **Upper Bound:** 122.50
- **Outliers Identified:** [150, 200, 250, 500]
- **After Capping - Min:** 25.00 — **Max:** 122.50

3 Problem 3: Choose the Right Method

Dataset A (Exam Scores)

Histogram & Decision: The data is highly concentrated near 100 with a few extreme lows (15, 20). This forms a **Left-Skewed** distribution. Because Z-Score relies on mean and standard deviation—which are distorted by skewed data—the **IQR Method** is the better choice for Dataset A.

Dataset B (House Prices)

Histogram & Decision: Prices are clustered in the lower range but trail off to extremely high values (1200). This represents a **Right-Skewed** distribution. Again, **IQR Method** is preferred.

Python Code for Dataset B

```
1 d3b = np.array([30, 32, 35, 38, 40, 42, 45, 48, 50, 55, 60, 65, 70, 80,
2               90,
3               100, 120, 150, 200, 300, 400, 500, 800, 1000, 1200])
4 q1_3b, q3_3b = np.percentile(d3b, [25, 75])
5 iqr3b = q3_3b - q1_3b
6 lb3b, ub3b = q1_3b - 1.5 * iqr3b, q3_3b + 1.5 * iqr3b
7
8 outliers3b = d3b[(d3b < lb3b) | (d3b > ub3b)]
```

Results (Dataset B)

- IQR Upper Bound: 432.50
- Outlier Count: 4
- Outliers: [500, 800, 1000, 1200]

4 Problem 4: Winsorization (Percentile Method)

Dataset (Ratings): [1, 2, 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 10, 1, 2, 10, 10, 10, 0.5, 9.5]

Python Code

```
1 d4 = np.array([1, 2, 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 10,
2               1, 2,
3               10, 10, 10, 0.5, 9.5])
4 # Calculate Percentiles (x = 5%)
5 p5_4 = np.percentile(d4, 5)
6 p95_4 = np.percentile(d4, 95)
7
8 # Winsorize data using numpy.clip
9 d4_capped = np.clip(d4, p5_4, p95_4)
```

Results & Justification

- **5th Percentile:** 1.00 — **95th Percentile:** 10.00
- **New Min:** 1.00 — **New Max:** 10.00

Why Winsorization is better here: Deleting extremes from survey/rating data leads to artificial bias, ignoring genuine customer satisfaction/dissatisfaction. Winsorization bounds the impact of severe anomalies without losing the sample entirely.

5 Problem 5: Mixed Methods – Compare Z-Score & IQR

Dataset: [12, 15, 18, 20, 22, 22, 23, 24, 25, 25, 26, 27, 28, 30, 32, 35, 38, 40, 45, 50, 55, 60, 65, 70, 80, 90, 100, 120, 150, 200, 500, 1000, 2000]

Python Code

```
1 d5 = np.array([12, 15, 18, 20, 22, 22, 23, 24, 25, 25, 26, 27, 28, 30,
2               32,
3               35, 38, 40, 45, 50, 55, 60, 65, 70, 80, 90, 100, 120,
4               150, 200,
5               500, 1000, 2000])
6
7 # Z-Score Method
8 z_scores5 = (d5 - np.mean(d5)) / np.std(d5, ddof=1)
9 outliers5_z = d5[np.abs(z_scores5) > 3]
10
11 # IQR Method
12 q1_5, q3_5 = np.percentile(d5, [25, 75])
13 iqr5 = q3_5 - q1_5
14 lb5, ub5 = q1_5 - 1.5 * iqr5, q3_5 + 1.5 * iqr5
15 outliers5_iqr = d5[(d5 < lb5) | (d5 > ub5)]
```

Results & Justification

- **Histogram observation:** Strongly Right-Skewed.
- **Z-Score Outliers ($|Z| > 3$):** 1 outliers: [2000]
- **IQR Outliers:** 4 outliers: [200, 500, 1000, 2000]

Analysis: The IQR method found more outliers. Massive extreme values (1000, 2000) dramatically inflated the standard deviation, stretching the Z-Score bounds so wide that they “masked” the smaller outliers. We must **trust IQR** here since it relies on percentiles, immune to standard deviation distortion.

6 Problem 6: Real-World Scenario – Employee Bonus

Dataset (Bonus in k BDT): [5, 5, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 10, 10, 10, 11, 12, 13, 15, 18, 20, 25, 30, 35, 40, 50, 60, 80, 100, 120, 150, 200, 300]

Python Code

```
1 import pandas as pd
2
3 d6 = np.array([5, 5, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 10, 10, 10, 11, 12,
4               13,
5               15, 18, 20, 25, 30, 35, 40, 50, 60, 80, 100, 120, 150,
6               200, 300])
7
8 # Central Tendency
9 mean6 = np.mean(d6)
10 median6 = np.median(d6)
11 mode6 = pd.Series(d6).mode().tolist()
```

```

10
11 # IQR Outliers
12 q1_6, q3_6 = np.percentile(d6, [25, 75])
13 ub6 = q3_6 + 1.5 * (q3_6 - q1_6)
14 outliers6_iqr = d6[d6 > ub6]
15
16 # Winsorization (x = 3%)
17 p3_6 = np.percentile(d6, 3)
18 p97_6 = np.percentile(d6, 97)
19 d6_capped = np.clip(d6, p3_6, p97_6)
20 mean6_new = np.mean(d6_capped)

```

Results & Justification

- **Mean:** 42.24 — **Median:** 12.00 — **Mode:** [7, 8, 10]
- **Observation:** The mean is significantly higher than the median and mode, pulled up by massive executive bonuses (positive skew).
- **IQR Outliers:** [100, 120, 150, 200, 300]
- **Winsorization ($x = 3\%$):** 3rd Percentile: 5.00 — 97th Percentile: 204.00
- **New Min:** 5.00 — **New Max:** 204.00
- **New Mean:** 39.33

Is the new mean more reasonable? Yes. By limiting the impact of extreme bonuses (which skew the average heavily), the Winsorized mean of 39.33 provides a much more realistic baseline for typical employee compensation and HR policy making.